

Project: Country Explorer

Project Title

Country Explorer : Country Information and Capital Weather Application

Objective

The objective of this project is to develop a Python application that retrieves information about a country and displays the current weather conditions of its capital city using live APIs. The project demonstrates API integration, HTTP requests, JSON data processing, and formatted console output.

Technologies Used

- Python 3
 - Requests Library
 - API Ninjas Country API
 - Open-Meteo Weather API
 - Visual Studio Code
 - Git & GitHub
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Project Description

Country Explorer is a console-based Python application that allows users to enter the name of any country. The application retrieves real-time information about the selected country, including its capital city, region, population, and currency.

After obtaining the country information, the application uses the capital city to retrieve the current weather conditions, including temperature, humidity, and wind speed.

The project demonstrates how multiple APIs can be integrated into a single Python application to provide useful information in a simple and user-friendly format.

Features

- Accepts country name as user input.
 - Displays the capital city.
 - Displays the geographical region.
 - Displays the population.
 - Displays the official currency.
 - Displays current temperature.
 - Displays current humidity.
 - Displays current wind speed.
 - Handles invalid country names appropriately.
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Implementation Process

1. The user enters the name of a country.
 2. The application sends an HTTP request to the API Ninjas Country API.
 3. The API returns country information in JSON format.
 4. The application extracts the required details such as capital, region, population, and currency.
 5. Using the capital city, the application requests live weather information from the Open-Meteo API.
 6. The weather API returns the current weather details.
 7. The application displays all retrieved information in a formatted console output.
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APIs Used

API Ninjas Country API

Used for retrieving:

- Country Name
- Capital City
- Region
- Population
- Currency

Open-Meteo API

Used for retrieving:

- Current Temperature
- Humidity
- Wind Speed

Output

The application successfully displays:

- Country Information
- Capital City
- Region
- Population
- Currency
- Current Weather of the Capital City

(Output screenshots are included in the Screenshots folder.)

Conclusion

The Country Explorer project successfully demonstrates the integration of multiple REST APIs within a Python application. It provides real-time country information and weather details while showcasing practical concepts such as HTTP requests, JSON parsing, user input handling, and console-based application development. The project enhanced understanding of API integration and strengthened Python programming skills.